

Algorithm and Eve: How AI will impact women at work

6 December 2024

Key findings

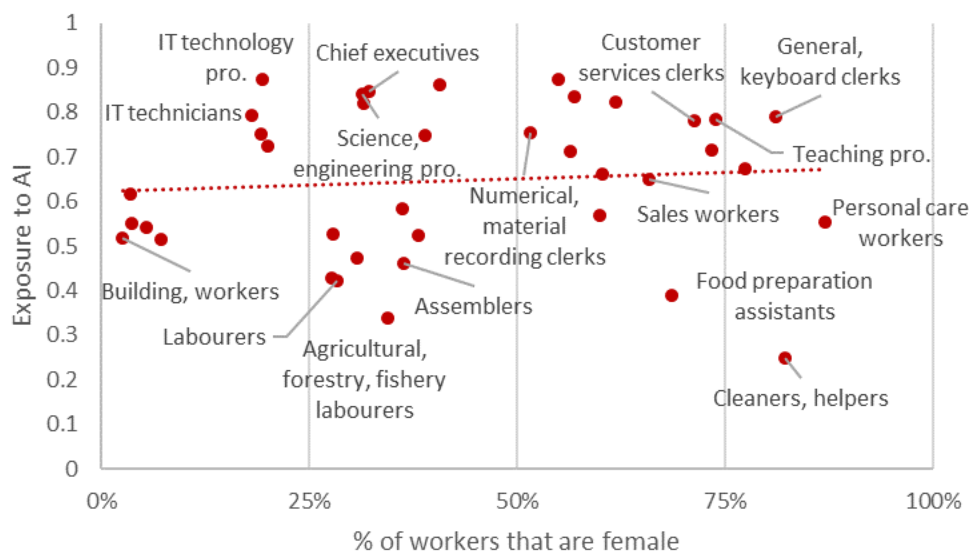
- Female and male workers face roughly the same occupational exposure to AI overall. However,
 - Women are underrepresented in the occupations with the very highest exposure to AI (e.g. science and engineering professional, chief executive).
 - Clerical occupations are not only characterised by high exposure to AI, but by an overrepresentation of women and particularly those without tertiary education.
- Employment growth between 2012 and 2022 was more rapid in those occupations most exposed to AI. Women's employment growth was even higher than men's in occupations highly exposed to AI, reflecting women's entry into traditionally male-dominated occupations.
- Women are still underrepresented in the AI workforce (the narrow set of workers with the skills to develop and maintain AI systems), among AI users (a broader category capturing workers who say that they interact with AI at work in one way or another) and among ICT graduates. Women report less positive perceptions about AI than men.
- This policy brief puts forward policy options to ensure that women and men alike can benefit from AI at work, including: applying a gender lens when monitoring AI's impact; following an inclusive approach to upskilling and reskilling; bridging gender divides in tech; combatting AI-induced bias; and using AI to combat bias.

In a recent study, female workers were 20 percentage points less likely to say they had used ChatGPT than male workers in the same occupation. While ChatGPT is just one AI tool in a rapidly evolving market, the finding raises questions about how women's and men's experiences of AI at work could differ. This is the question this policy brief aims to address, drawing from the OECD working paper "Who will be the workers most affected by AI?" (Lane, 2024^[1]). The policy brief explores the gender composition of occupations highly exposed to AI and assesses women's access to AI-related employment opportunities and to productivity-enhancing AI tools. It concludes with a set of policy options that policymakers could pursue to ensure that women and men alike can benefit from AI at work.

Are men or women more exposed to AI at work?

Female and male workers face roughly the same occupational exposure to AI overall, as shown by the relatively flat trend line in Figure 1. Occupational exposure to AI refers to the degree of overlap between the abilities required in an occupation and the technical abilities of AI. It is well established that due to AI's progress in automating non-routine, cognitive tasks, the occupations most exposed to AI tend to be “white collar occupations” typically requiring several years of formal training and/or tertiary education. Occupations with the highest exposure to AI will be most impacted by AI, most exposed to the associated risks and opportunities, and could face the most disruption.

Figure 1. Female and male workers face roughly the same occupational exposure to AI overall



Note: Non-weighted averages over 22 countries for which data are available: Austria, Belgium, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Lithuania, the Netherlands, Norway, Poland, the Slovak Republic, Slovenia, Spain, Sweden, the United Kingdom and the United States.

Source: Lane (2024^[1]), “Who will be the workers most affected by AI? A closer look at the impact of AI on women, low-skilled workers and other groups”, <https://doi.org/10.1787/14dc6f89-en> using 2022 data from EU-LFS, US-CPS and UK-LFS.

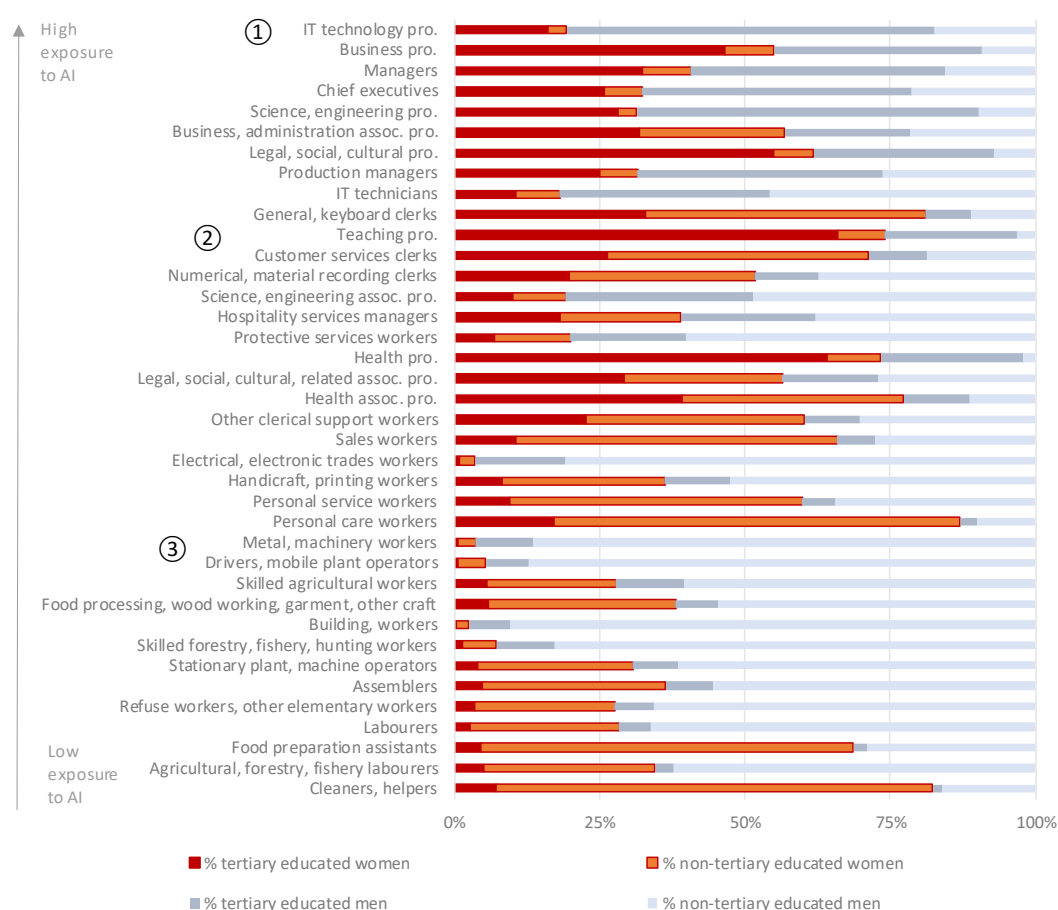
The markers in Figure 2 highlight three points of relevance for policymakers who wish to assess and monitor the different risks and opportunities applicable to men and women in different occupations in their countries:

1. **The underrepresentation of women in the occupations with the very highest exposure to AI and whether this could hold women back from opportunities associated with AI.** The occupations most exposed to AI (shown beside the ① marker in Figure 2, which orders occupations from high to low exposure to AI) tend to be male-dominated occupations that typically require tertiary education. For instance, the occupations of IT technology professional, chief executive and science and engineering professional all have under 35% female representation, although women make up more than half of business professionals, also very highly exposed.
2. **The high representation of women in clerical occupations, which could be at particular risk of automation by recent (and future) advances in generative AI,** as suggested by Gmyrek, Berg and Bescond (2023^[2]). The highly exposed occupations shown beside the ② marker in Figure 2 have a high representation of women and do not typically require tertiary education.

Examples include: general, keyboard clerk; customer service clerk; and numerical recording clerk. The relevance of tertiary education is that some studies have shown that AI exposure is linked to more positive outcomes among more educated workers (as discussed in the next section). Teaching professional, another female-dominated occupation, has a similar level of exposure to AI, although most teaching professionals do have tertiary education.

3. **The men without tertiary education in occupations which are not highly exposed to AI but are at high risk of automation (from all technologies).** Examples appear beside and below the ③ marker in Figure 2 and include professions typically associated with trades (e.g. metal, machinery worker; driver, mobile plant operator; building worker) as well as occupations which rely heavily on manual skills and strength (such as labourers and refuse worker, other elementary workers). Previous OECD work (Lassébie and Quintini, 2022^[3]) shows that men are more likely to hold jobs at higher risk of automation (i.e. from all technologies, not just AI) than women.

Figure 2. Women are underrepresented in the occupations with the very highest exposure to AI



Note: Non-weighted averages over 22 countries for which data are available: Austria, Belgium, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Lithuania, the Netherlands, Norway, Poland, the Slovak Republic, Slovenia, Spain, Sweden, the United Kingdom and the United States.

Source: Lane (2024^[1]), "Who will be the workers most affected by AI? A closer look at the impact of AI on women, low-skilled workers and other groups", <https://doi.org/10.1787/14dc6f89-en> using 2022 data from EU-LFS, US-CPS and UK-LFS.

What are the implications of high exposure to AI?

Occupations with the highest exposure to AI will be most impacted by AI and could face the most disruption. However, the empirical literature to date (as summarised in the 2023 OECD Employment Outlook (2023^[4])) provides little evidence of negative employment outcomes due to AI. Some studies even suggest that AI exposure is linked to positive outcomes and that these links are stronger among more educated and higher-income workers. If the process of adapting to AI overwhelmingly favours more educated workers and higher-income workers, then AI will deepen existing inequalities.

New analysis (Lane, 2024^[1]) reinforces the idea of a positive relationship between AI exposure and employment. It also shows that women's employment growth in the period from 2012 to 2022 was even higher than men's in occupations highly exposed to AI (while controlling for other technological advances, offshorability and international trade as well as from trends at occupation and country levels). However, it is difficult to establish whether this is the impact of AI. Rather than suggesting that AI has created opportunities more suited to women, this difference may reflect declining gender segregation within traditionally male-dominated occupations typically requiring tertiary education. In other words, despite the underrepresentation of women in the occupations with the very highest exposure to AI (at the top of Figure 2), female representation grew in many of these occupations between 2012 and 2022, e.g. from 25% to 32% among chief executives and from 27% to 31% among science and engineering professionals.

While employment growth is a useful indicator of underlying labour demand in occupations exposed to AI, this is just one of the ways in which AI is expected to change the workplace. To date, AI may have even impacted job quality more than job quantity. In the OECD AI surveys of employers and workers (Lane, Williams and Broecke, 2023^[5]), which were conducted in early 2022, workers using AI in finance and manufacturing were generally very positive about the impact of AI on their working conditions. Many workers reported that their tasks had been reorganised while employers confirmed that skill needs were changing. Many workers said that they worked at a faster pace due to AI while others expressed concerns about excessive data collection.

Furthermore, how AI impacts exposed occupations may depend on the type of AI being used (for instance, the impact of generative AI will be different to the impact of computer vision) or the characteristics or the context of the user. Much of the analysis in this domain relies on occupational AI exposure measures, which can only pick up differences between occupations, not within. For instance, it cannot distinguish between a female scientist in an entry-level role in a manufacturing plant and her more experienced male colleague, even if there are differences in the nature of their jobs, their interaction with AI, their attitudes towards AI, and even their employers' perceptions of their ability to use AI. The rest of the policy brief moves away from the occupational AI exposure measures to examine some of these issues, in which gender can play a role.

Do women have less access to AI-related opportunities at work?

Women have less access to AI-related employment opportunities and to productivity-enhancing AI tools in the workplace, which could prevent the benefits of AI from being broadly and fairly shared. This is based on recent OECD studies showing that women are underrepresented in the AI workforce (the narrow set of workers with the skills to develop and maintain AI systems) and among AI users (a broader category capturing workers who say that they interact with AI at work in one way or another).

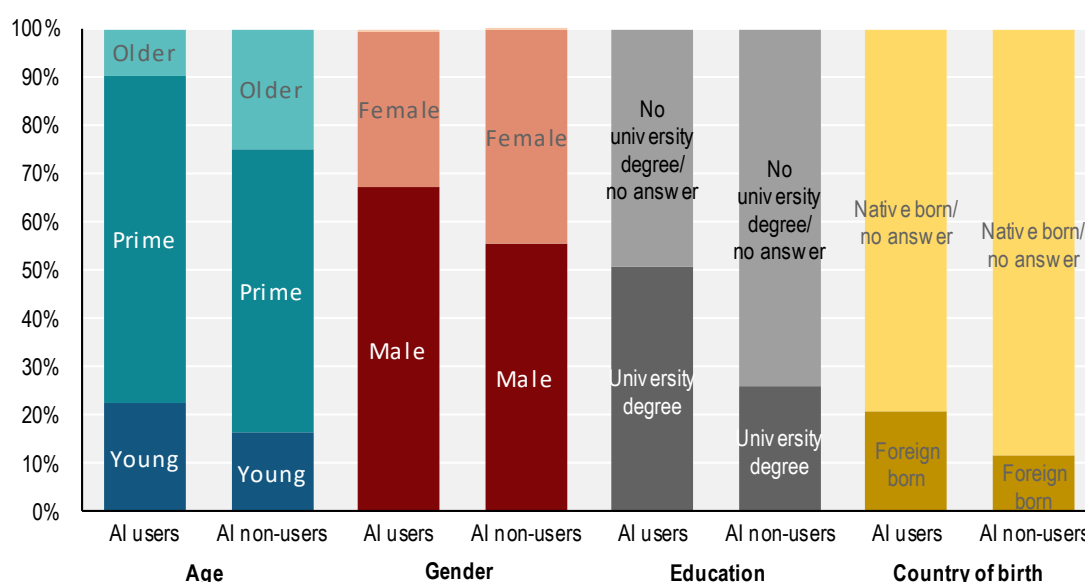
Green and Lamby (2023^[6]) examine the "AI workforce", defined as those with the skills to develop and maintain AI systems. The authors find that the AI workforce is small – accounting for just above 0.3% of employment across OECD countries – and confined to a narrow socio-demographic segment of the population, primarily male and university educated. As many ICT- and AI-related jobs are relatively lucrative, women could find themselves disadvantaged in terms of employment opportunities. Additionally,

underrepresentation of women (or of any socio-demographic group) in decision-making roles developing and implementing AI increases the risk that the experiences and voices of this group are omitted from the process.

The OECD AI survey of workers revealed similar socio-demographic patterns among AI users, defined as those who say that they interact with AI at work in one way or another. In both sectors, AI users were more likely to be younger, male and university educated compared to non-users (Figure 3). 41% of male workers surveyed were AI users compared to 29% of women. It is worth noting that most AI users had positive views on AI's impact on their performance and working conditions, with the implication that groups less likely to use AI are at a perceived disadvantage and risk being excluded from the opportunities that AI use can bring.

Figure 3. AI users are more likely to be younger, male, foreign-born and have a university education than non-users

Percentage of workers in companies that use AI, by age, gender and education



Note: Graph shows simple average across workers in the manufacturing sector and workers in the financial sector. Workers in companies that use AI were asked: "Which of these statements best describes your interaction with AI at work? I work with AI; I manage workers who work with AI; I develop/maintain AI; I am managed by AI; I interact with AI in another way; I have no interaction with AI at work; Don't know". Workers who selected "no interaction" or "Don't know" or who said that their companies did not use AI are described as "non-users", while the rest are described as "AI users".

Source: OECD worker survey on the impact of AI on the workplace (2022).

Furthermore, the same survey showed that male AI users were more likely than female AI users to say that AI had improved their productivity and working conditions, that they had specialised AI skills, and that they were enthusiastic to learn more about AI. Men were more likely to say that they expected AI to increase wages in their sector while women were more likely to say that they were very or extremely worried about losing their jobs in the next 10 years. The gender differences were statistically significant even when controlling for age, sector, occupation and overall sentiment about technology. For instance, male AI users were more likely to be managers and professionals (whose views about AI were typically more positive), while female AI users were more likely to be clerical support or service and sales workers (whose views were typically less positive), but occupation only explained a quarter of the gender gap regarding performance and working conditions. In other words, male managers and professionals were

still more positive than female managers and professionals about the impact of AI on performance and working conditions.

These findings are in line with the Danish study referenced in the introduction (Humlum and Vestergaard, 2024^[7]), in which female workers were 20 percentage points less likely to say they had used ChatGPT than male workers in the same occupation. Most of this gender gap persists (it reduces to 17 percentage points) when comparing workers within the same workplace and controlling for workers' detailed task mixes. According to the authors, the best explanation for this gender gap lies in the tendency for women to report that they needed training before using ChatGPT.

While these surveys rely on attitudes, expectations and self-reporting, it is plausible that the greater confidence and more positive perceptions of AI among male workers could be an advantage when it comes to engaging with AI, gaining relevant skills and ultimately benefiting from opportunities related to AI.

Where do these gender divides originate?

Uneven access to opportunities associated with AI may reflect digital skill divides as well as women's lower participation in science-related tertiary education. A new OECD report on Bridging talent shortages in tech (OECD, 2024^[8]) documents the gender divide in technology and puts forward a few possible explanations, which likely work in combination:

- Stereotyping seems to take hold at a young age whereby, at some point in their schooling, girls become discouraged from scientific careers despite showing similar aptitudes as boys.
- A lack of role models in STEM has been shown to have an impact on girls' subject choices in school and university.
- Discrimination in recruitment processes in technology and STEM means that women are less likely to gain employment even when they have the skills.
- Unfavourable working conditions in technology and STEM could discourage women from these sectors.

Addressing these gender divides is crucial to guarantee that women of all ages can participate fully in the digital economy.

What policy options can be pursued to ensure that women and men can benefit from AI at work?

This policy brief puts forward the following policy options to ensure that women and men alike can benefit from AI at work:

- **Apply a gender lens when monitoring AI's impact:** As new AI advances emerge and as firms' AI use broadens and matures, policymakers and researchers will want to monitor AI's impact on employment outcomes through a gender lens. Groups facing disproportionate harm and requiring support could include non-tertiary-educated female workers in clerical occupations highly exposed to AI and non-tertiary-educated male workers in occupations at high risk of automation.
- **Follow an inclusive approach to upskilling and reskilling:** Policymakers will want to equip workers, female and male, with the right skills so they are empowered to work with AI, adapt to changes on the job, or move from declining sectors and occupations into to new and growing ones. Reskilling and upskilling workers is crucial to seize the benefits of AI and to ensure a fair transition for workers, and this should be done in an inclusive and accessible manner.

- **Bridge gender divides in tech:** An additional targeted effort may be needed to bridge the divides that currently hold women back from opportunities associated with AI, for instance: imbedding inclusivity into school curricula and teacher training; following a skills-first hiring approach and mandating diversity on recruitment panels; developing metrics for employers to track diversity and performance; building an inclusive work culture; providing flexible work options; supporting women-led tech businesses; and promoting role models for women in the tech sector.
- **Combat AI-induced bias:** Policymakers must ensure that existing anti-discrimination legislation is suitable for governing AI systems and must be vigilant about the potential for gaps or loopholes to be exploited. Poorly designed or biased AI systems, trained on selective and insufficiently diverse data, can amplify labour market biases, including gender biases.
- **Use AI to combat bias:** AI itself may offer solutions for opening up new opportunities for traditionally underrepresented groups, by identifying human bias and discrimination and offering new data-driven methods to inform decision-making. The EU AI Act calls on EU member countries to support and promote research and development of AI solutions in support of socially beneficial outcomes, such as AI-based solutions to tackle socio-economic inequalities.

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Explore further

Read the related working paper:

Lane, M. (2024), “Who will be the workers most affected by AI?: A closer look at the impact of AI on women, low-skilled workers and other groups”, *OECD Artificial Intelligence Papers*, No. 26, OECD Publishing, Paris, <https://doi.org/10.1787/14dc6f89-en>.

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This policy brief contributes to the OECD’s Artificial Intelligence in Work, Innovation, Productivity and Skills (AI-WIPS) programme, which provides policymakers with new evidence and analysis to keep abreast of the fast-evolving changes in AI capabilities and diffusion and their implications for the world of work. The programme aims to help ensure that adoption of AI in the world of work is effective, beneficial to all, people-centred and accepted by the population at large. AI-WIPS is supported by the German Federal Ministry of Labour and Social Affairs (BMAS) and will complement the work of the German AI Observatory in the Ministry’s Policy Lab Digital, Work & Society. For more information, visit <https://oecd.ai/work-innovation-productivity-skills> and <https://denkfabrik-bmas.de/>.



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